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SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes

Total Marks: 25

Code of Conduct

I Sahana 192071023 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Sahana

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1	6	4	3	4	2	19
Q2	6	4	3	4	2	19
Q3	6	4	3	4	2	19
Q4	6	4	3	4	2	19
Q5	5	4	3	4	2	18

S. Sahana
Signature of the Faculty

Total Marks Awarded:

94

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Explain the Evolution of cloud computing

Cloud computing Evolved through several stages of technical development

Initially standalone computers where the service and the storage are given in the single Machine.

Network computing Emerged allowing computers to communicate through Local Area Network (LAN) and wide Area Network.

① Traditional IT

→ The oldest computing Model where the organisation or user owns and manages the Entire IT

The user is responsible for the purchasing, hardware, installing operating system, maintaining server, storing data, Installing software, ensuring security.

→ It is Expensive time consuming and difficult to scale whenever business requirement increases.

Ex: College buying its own server, networking equipment, software licenses and maintaining everything in computer lab.

② Infrastructure as a service (IaaS)

IaaS is the first stage of cloud computing. In this model the cloud provider supplies Virtualized computer resource such as servers, storage, networking and Virtualization

IaaS provides flexibility because users can rent computing resources whenever needed and pay only for what they use. Suitable for organisations that want complete control over their software, avoiding cost of physical hardware.

Ex: Amazon EC2 (AWS), Microsoft Azure Virtual Machine, compute Engine.

③ Platform as a service (PaaS)

PaaS allows developer to concentrate on software development without worrying about server configuration or maintenance. It speeds up application development.

- Faster application development
- No server maintenance
- Built in development tools
- Automatic updates

Ex: Google App Engine, Microsoft Azure App service, Heroku.

④ Software as a service (SaaS)

Software as a service delivers complete software application over the internet simply access the application using a web browser or mobile app.

The cloud provider manages the update, security, backup and maintenance.

Ex: Gmail, Google docs, Microsoft 365, Zoom, Salesforce.
Customer Manages
Data (files, user information, settings)

Everything as a service (XaaS)

organization can choose only the service they need without investing in expensive infrastructure, XaaS provides maximum flexibility, scalability and cost efficiency.

SaaS: Gmail, Microsoft 365

PaaS: Heroku, Google app Engine

IaaS: AWS EC2

Q2) Explain Hypervisors and compare Type 1 and Type 2 hypervisors.

Hypervisor is a software layer that enables virtualization by creating and managing multiple virtual machines on a single physical computer.

It acts as an intermediary between the hardware and the virtual machine, allocating resources such as CPU, memory, storage, network bandwidth.

There are two main types of hypervisor

- ① Hypervisor (Bare-Metal hypervisor)
- ② Hypervisor (Hosted Hypervisor)

Base - Metal Hypervisor

Run directly on the physical hardware without required a host operating system.

It provides better performance, efficiency, security.

commonly used in cloud data centre and enterprise environment.

VMware ESXi, Microsoft Hyper-V, Xen server

Hosted Hypervisor

Top on existing operating system such as windows or Linux. It relies on the host operating system access hardware resources.

Easier to install and use. it generally offer lower performance compared to type 1 hypervisor.

Q3) Explain the role of API in cloud computing

→ An API is a set of rules that allows different software application to communicate.

Acts as a bridge btw cloud service and application.

Role of API cloud computing

→ Enable communication btw cloud service and client application.

→ Allows or create, update, delete, manage the cloud resource.

→ Helps to integrate multiple cloud service.

→ ready to use cloud function.

How API support interoperability

- connect application developed in different programming language
- Enable communication between different cloud platforms
- Allows data sharing between multiple services.
- support seamless integration and third party services

Ex:

Google Map API

AWS EC2 API

Paypal Payment API

weather API

Q4) Explain resource pooling in cloud computing

Resource Pooling

Resource pooling is a cloud computing feature where computing resource shared among multiple user

Resource include servers, storage, memory, CPU and networking

Resource are dynamically allocated and reallocated based on user demand.

How it works.

- All the physical resource are grouped into a common pool
- user request resources whenever needed
- The cloud provider automatically assigns available resource.

Benefits

- Efficient utilization hardware resource.
- Reduces Infrastructure & Maintenance cost
- Provides better scalability
- Improves resource availability
- Ensure secure isolation
- Amazon web service
- Microsoft Azure
- Google cloud platform

15) Explain rapid elasticity in cloud computing

Rapid Elasticity

- Rapid Elasticity is the ability of cloud computing to automatically increase or decrease resource based on demand.

Resource can be Scaled up during high workloads and Scaled down when demand decreases

- The cloud continuously monitor resource usage
- Additional resource are automatically allocated during peak demand
- Scaling happens quickly without interrupting user.

Benefits

- Handle sudden increase in workload
- Improves application performance
- High availability
- Provides better user experience.
- Supports business growth.
- Pay only resources used.

Ex

- Amazon Great Indian Festival
- Flipkart big billion days
- Netflix increasing server during weekend
- online ticket booking during festival.